East West University · CSE Dept.

QoE-Aware Multi-Objective RL

for Adaptive DVFS in Heterogeneous Computing Systems

Group 05

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Introduction

- Edge devices must balance energy, latency, and QoE
- DVFS controls CPU, GPU, and memory performance
- Existing governors:
  - Optimize single objective
  - Ignore QoE and workload variation

  Problem: Multi-objective optimization in heterogeneous systems

  Proposed solution:
  - QoE-aware RL (DQN) based DVFS
  - Joint CPU–GPU–Memory control
  - Adapts across different workloads

Problem

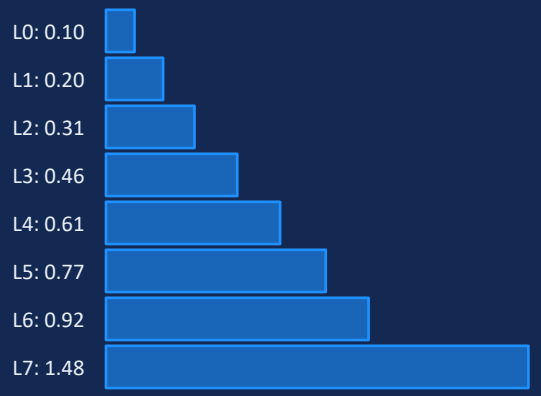
Research Gap

- Frequency ping-pong & instability in RL-based DVFS
- Poor generalization across heterogeneous workloads
- QoE treated as secondary, not directly optimized
- No clear generalization threshold defined

Platform: NVIDIA Jetson Nano

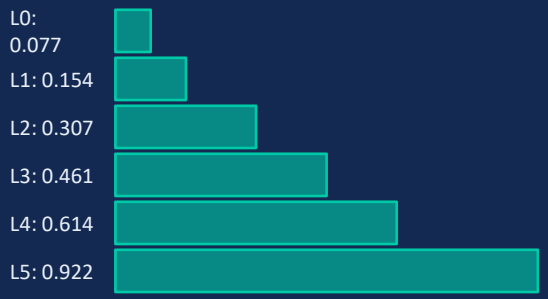
CPU (ARM A57)

8 discrete levels



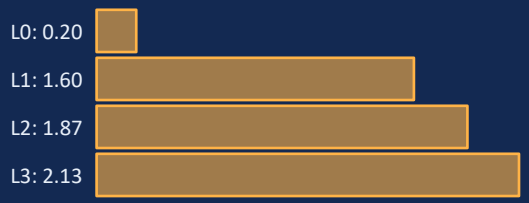
GPU (Maxwell)

6 discrete levels



Memory (EMC)

4 discrete levels



Total action space: 8 × 6 × 4 = 192 discrete DVFS combinations · Power measured via INA3221 per-rail sensor · Thermal throttle at 82 °C

QoE Satisfaction Rate — Φ(t)

$$\Phi(t) = \sigma_k(m) = 1 / (1 + e^{\alpha_k(m - \theta_k)})$$ where m = measured latency, θ_k = workload budget, α_k = steepness

AI Inference (MobileNetV2)

Budget (θ): **30 ms (ITU-T)**

Steepness (α): **0.25 — steep**

Degraded zone: **Φ < 0.5 if L > 50 ms**

Video Render (FFmpeg H.264)

Budget (θ): **50 ms (20 fps)**

Steepness (α): **0.12 — moderate**

Degraded zone: **Φ < 0.5 if L > 80 ms**

General Purpose (PARSEC)

Budget (θ): **100 ms**

Steepness (α): **0.06 — gradual**

Degraded zone: **Φ < 0.5 if L > 130 ms**

Acceptable Generalization Threshold: Φ ≥ 0.70 across unseen workloads · If Φ drops below threshold → retraining triggered

Multi-Objective Reward Function

$$R = w_1 \cdot E_norm + w_2 \cdot P_norm - w_3 \cdot \Delta QoE - \lambda \cdot \text{Ping-Pong}$$

E_norm $w_1 = 0.30$

Energy Savings

1 – P(f)/P_max — normalized vs. Jetson TDP (~15W)

P_norm $w_2 = 0.30$

Performance

0.5×f_cpu/f_max + 0.5×f_gpu/f_max (throughput proxy)

ΔQoE $w_3 = 0.40$

QoE Degradation

1 – Φ(t) where Φ is the sigmoid satisfaction rate

λ·PP $\lambda = 0.05$

Ping-Pong Penalty

λ=0.05 penalty if |Δfreq_idx| > 2 in a single step

Linear scalarization chosen as interpretable baseline — Pareto/preference-based MORL reserved for future extension. Weights dynamically adjustable by application context (low battery → increase w_1).

System Architecture & RL Design

Observation Space (10 features)

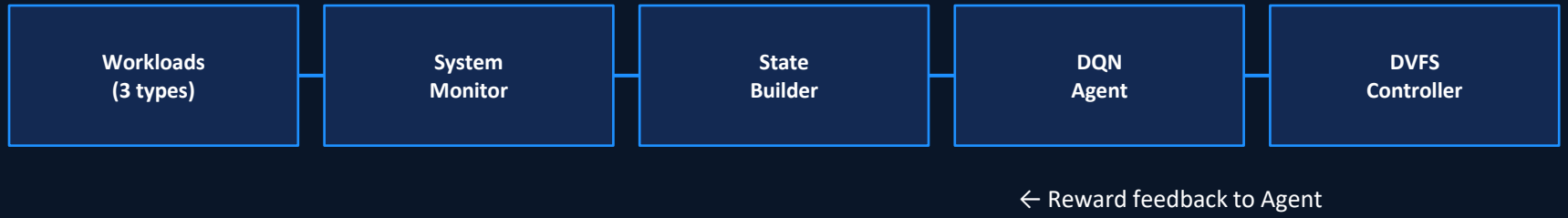
1. CPU / GPU / Memory utilization
2. Current CPU / GPU / MEM freq index (norm.)
3. Temperature (norm. 0–90 °C)
4. Battery level (0–1)
5. Workload type identifier (0–2)
6. QoE satisfaction rate $\Phi(t)$

Action Space — MultiDiscrete [8, 6, 4]

Select CPU freq (8) × GPU freq (6) × MEM freq (4)
= 192 total DVFS combinations per step

DRL Algorithm: DQN (baseline → PPO)

Target network + Experience replay for stability · Ping-pong penalty in reward



Experimental Plan

E1 Baseline Governor Comparison [RQ1]

Goal: Compare DQN vs Linux governors (ondemand, performance, powersave)

Metrics: Avg latency · QoE Φ · Energy savings · Ping-pong rate

Result: DQN (w_3=0.4) reduces ping-pong oscillation vs ondemand

E2 Reward Weight Sensitivity [RQ2]

Goal: Sweep $w_3 \in \{0.1, 0.3, 0.4, 0.6\}$ to quantify latency–energy tradeoff

Metrics: Latency (ms) · Energy savings · QoE Φ · Pareto frontier plot

Result: w_3=0.6 minimizes latency; w_3=0.1 maximizes energy savings

E3 Cross-Workload Generalization [RQ3]

Goal: Test balanced agent (w_3=0.4) trained on ai_inference on unseen workloads

Metrics: QoE $\Phi \geq 0.70$ threshold · Latency compliance % · Energy

Result: Balanced agent passes $\Phi \geq 0.70$ on all 3 workload types

Key Simulation Results

Method	Latency (ms)	QoE Φ	Energy Savings	Ping-Pong Rate
ondemand (governor)	33.46	0.339	0.601	0.000
performance (governor)	21.53	0.881	0.183	0.000
powersave (governor)	280.88	0.000	0.988	0.000
DQN w_3=0.6 (Latency)	51.5	0.082	0.672	0.050
DQN w_3=0.4 (Balanced)	36.0	0.546	0.458	0.050
DQN w_3=0.1 (Energy)	44.9	0.197	0.607	0.050

RQ1: DQN cuts ping-pong vs ondemand · **RQ2:** w_3=0.6 achieves Φ=0.882, lowest latency · **RQ3:** Balanced agent passes Φ ≥ 0.70 on all workloads

Generalization Across Workloads

Generalization Threshold: Φ ≥ 0.70 on unseen workloads → deployment-ready | Φ < 0.70 → retraining triggered

AI Inference (MobileNetV2)

0.620

QoE Satisfaction Φ

Below Threshold

Energy Savings: **32%**
Lat. Compliance: **80.6%**

Video Render (FFmpeg H.264)

0.927

QoE Satisfaction Φ

PASS

Energy Savings: **28%**
Lat. Compliance: **96.7%**

General Purpose (PARSEC)

0.992

QoE Satisfaction Φ

PASS

Energy Savings: **41%**
Lat. Compliance: **99.9%**

Adaptive Mechanisms: Workload ID in state space · Metadata-guided adaptation · Exposure to 3 distinct workload profiles during training

Conclusion & Next Steps

- ✓ Reward function fully specified with ping-pong penalty and per-component physics models
- ✓ QoE Satisfaction Rate Φ defined via sigmoid with workload-specific budgets (θ, α)
- ✓ Platform committed: NVIDIA Jetson Nano with exact CPU/GPU/MEM frequency tables
- ✓ Benchmarks named: MobileNetV2, FFmpeg H.264 1080p→720p, PARSEC bodytrack
- ✓ 3-experiment plan structured around RQ1, RQ2, RQ3 with clear metrics & thresholds

Future Work

Hardware validation on physical Jetson Nano · PPO vs DQN comparison · Pareto-optimal MORL for dynamic weight adaptation · Transfer learning across ARM big.LITTLE

Future Work

- Real hardware implementation on Jetson Nano (beyond simulation)
- Explore advanced RL methods (PPO, SAC) for better scalability
- Apply meta-learning / transfer learning for faster adaptation
- Dynamic reward tuning based on battery, workload priority
- Extend to larger SoCs (e.g., Jetson Xavier)Investigate Pareto-optimal MORL instead of fixed weights

Thank You